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# Forty Years of Reading Intervention Research for Elementary Students with or at Risk for Dyslexia: A Systematic Review and Meta-Analysis

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## ABSTRACT

This meta-analysis included experimental or quasi-experimental intervention studies conducted between 1980 and 2020 that aimed to improve reading outcomes for Grade K-5 students with or at risk for dyslexia (i.e., students with or at risk for word reading difficulties, defined as scoring at or below norm-referenced screening or mean baseline performance thresholds articulated in our inclusion criteria). In all, 53 studies reported in 52 publications met inclusion criteria ($m$ = 351; total student $N$ = 6,053). We employed robust variance estimation to address dependent effect sizes arising from multiple outcomes and comparisons within studies. Results indicated a statistically significant main effect of instruction on norm-referenced reading outcomes ($g$ = 0.33; $p < .001$). Because there was significant heterogeneity in effect sizes across studies ($p < .01$), we used meta-regression to identify the degree to which student characteristics (i.e., grade level), intervention characteristics (i.e., dosage, instructional components, multisensory nature, instructional group size), reading outcome domain (i.e., phonological awareness, word reading/spelling, passage reading, or reading comprehension), or research methods (i.e., sample size, study design) influenced intervention effects. Dosage and reading outcome domain were the only variables that significantly moderated intervention effects ($p$ = .040 and $p$ = .024, respectively), with higher dosage studies associated with larger effects ($b$ = 0.002) and reading comprehension outcomes associated with smaller effects than word reading/spelling outcomes ($b$ = −0.080).

Dyslexia is a specific learning disability characterized by difficulties with accurate or fluent word recognition and spelling (Fletcher et al., 2019). When operationalized as word reading accuracy more than 1.5 standard deviations below the mean, estimates of dyslexia's prevalence in the population are roughly 7% (Peterson & Pennington, 2012). Research in cognitive neuroscience, genetics, developmental psychology, and education has demonstrated that dyslexia is neurobiological in origin and heavily determined by genes, while also influenced by environmental factors (Fletcher et al., 2019; Grigorenko et al., 2020). Individuals with dyslexia are at greater risk for negative academic and occupational outcomes (Boscardin et al., 2008; Daniel et al., 2006; Hernandez, 2011). That said, early identification and remediation has been found to increase both the academic and emotional well-being of individuals with or at risk for developmental dyslexia, even demonstrating efficacy in preventing the incidence of reading difficulties (RDs) altogether (e.g., Al Otaiba et al., 2009; Mathes et al., 2005).

Although individual intervention studies and descriptive syntheses have furnished information about what works to improve outcomes for young students with dyslexia, there is a need for meta-analytic research

examining the effects of reading instruction on reading outcomes for this population specifically. Previous meta-analyses (e.g., Gersten et al., 2020; Neitzel et al., 2022; Slavin et al., 2011; Suggate, 2010; Swanson et al., 1999; Wanzek et al., 2016, 2018) report robust findings about effective early reading instruction for elementary-grade students with or at risk for RDs broadly defined (i.e., defined as including a wide range of reading and language difficulties). Specifically, these systematic reviews reveal that multicomponent reading interventions that provide explicit, systematic instruction in foundational skills (going forward, we use the term *foundational skills* to refer to phonological awareness [PA], phonics knowledge, word reading, spelling, and connected-text reading) and simultaneously focus on meaning (i.e., both word meanings and comprehension of connected text) are associated with significant positive effects. Three of the most recent among these meta-analyses, all of which employed stringent standards for inclusion and sophisticated meta-analytic methods, reported mean intervention effects of 0.23 (Neitzel et al., 2022) and 0.39 (Gersten et al., 2020; Wanzek et al., 2018).

It is possible and even likely that reading interventions that include an explicit, systematic approach to foundational skills instruction are effective, with effects estimated in the same range, for students with or at risk for dyslexia specifically. Individual intervention studies demonstrate that word RDs can be effectively remediated when instruction systematically and explicitly develops students' PA, teaches grapheme–phoneme correspondences, and provides practice reading words and connected text (e.g., Lovett et al., 2000; Morris et al., 2012; Vellutino et al., 2004). However, because no large-scale, rigorous meta-analysis focuses on the effects of reading instruction or intervention for this population of students, this hypothesis is difficult to confirm.

A synthesis of intervention research findings for students with or at risk for dyslexia depends on a clear operational definition of dyslexia. Miciak and Fletcher (2020) demonstrate a lack of validity for approaches to dyslexia identification that seek "to subdivide children into dyslexia and nondyslexic groups based on IQ, cognitive discrepancies, or other hypothesized markers of unexpectedness or etiology" (p. 7). Persuaded by their review of research, we defined dyslexia as the lower part of a continuous distribution of word reading and spelling skill (Miciak & Fletcher, 2020). We operationalized this definition by including studies that used a participant screening threshold of performance at or below the 25th percentile or reported mean baseline performance at or below the 16th percentile on a standardized measure of word reading, spelling, or skills foundational to word reading (e.g., PA, phonics knowledge). By focusing specifically on intervention effects on reading outcomes for this population of students, this review has the potential to demonstrate whether findings about effective instruction for students with RDs broadly hold for students with or at risk for dyslexia specifically.

# Recent Meta-Analyses of Reading Intervention Research

## *Participant Samples*

During the last two decades, seven meta-analyses (Donegan & Wanzek, 2021; Gersten et al., 2020; Neitzel et al., 2022; Slavin et al., 2011; Suggate, 2010; Wanzek et al., 2016, 2018; see Table 1) have investigated the immediate effects of reading instruction on word reading outcomes for elementary-grade students with or at risk for RDs. One additional research meta-analysis (Galuschka et al., 2020) investigated the immediate effects of spelling instruction on spelling outcomes for students with reading or spelling difficulties. Other rigorous reviews have been conducted as well, but they (a) focused only on comprehension outcomes (Denton et al., 2022; Hwang et al., 2021); (b) focused only on follow-up effects measured after the conclusion of the intervention (Suggate, 2016); (c) only included particular types of reading interventions (e.g., branded or unbranded Orton-Gillingham interventions; Ritchey & Goeke, 2006; Stevens et al., 2021); (d) included students in middle or high school (Flynn et al., 2012; Scammacca et al., 2015, 2016; Wanzek et al., 2013); (e) focused on a different or narrower subset of students, such as English learners (Cheung & Slavin, 2012; Ludwig et al., 2019; Richards-Tutor et al., 2016; Roberts, Hall, et al., 2021); or (f) primarily explored associations between student characteristics and intervention effects (Al Otaiba & Fuchs, 2002; Stuebing et al., 2015; Tran et al., 2011).

The seven meta-analyses that are most relevant to the current review (see Table 1) included studies with elementary-grade participants with or at risk for RDs who did *not* exhibit word reading or spelling difficulties (i.e., students with specific reading comprehension difficulties and language impairments), as well as including participants who did. The Wanzek et al. (2016) review included studies with participants in kindergarten (K) through Grade 3 who had "low achievement, low phonemic awareness, low income, language disorders" (p. 556). Grade K-5 students included in studies meta-analyzed by Slavin et al. (2011) were "children in the lowest 33% (or lower) of their classes, or any children receiving tutoring or other intensive services to prevent or remediate serious reading problems" (p. 5). Neitzel et al. (2022), who sought to replicate and extend the Slavin et al. meta-analysis, included Grade-6 "students reading below grade level or reading in the lowest half … of their grade" (p. 153). Donegan and Wanzek (2021) included Grade 4–5 students "described by study authors as below grade level in reading, at risk for



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**TABLE 1**
**Comparing Characteristics of Previous Meta-Analyses of Reading Intervention Research**

| | Suggate (2010) | Slavin et al. (2011) | Wanzek et al. (2016) | Wanzek et al. (2018) | Gersten et al. (2020) | Donegan and Wanzek (2021) | Neitzel et al. (2022) | Current meta-analysis |
|---|---|---|---|---|---|---|---|---|
| Search dates | 1982–2007 | 1970–2009 | 1995–2013 | 1995–2015 | 2002–2017 | 1988–2019 | 1990–2020 | 1980–2020 |
| Grey literature | N | Y | N | N | N | N | Y | Y |
| Total studies | 85 | 96 | 72 | 25 | 33 | 33 | 65 | 53 |
| Participant grades | PK-7 | K-5 | K-3 | K-3 | 1–3 | 4–5 | K-6 | K-5 |
| Included interventions | "A reading intervention" (p. 1560) | "Specific, potentially replicable programs for children who are having difficulties learning to read" (p. 5) | "Interventions targeted early literacy" (p. 556) | "Interventions targeted early literacy" (p. 614) | "A reading intervention: that is, preventative instructional practices and activities designed to help students who are considered at risk for reading difficulties" (p. 405) | "A reading intervention targeting literacy in English for students struggling with reading was provided" (p. 1947) | "A reading program, defined as a specific, replicable school day/school year approach … designed to improve the reading achievement of struggling readers" (p. 152) | "Instructio … focused on PA, phonics, word reading, spelling, or a combination of these domains. Studies that had other intervention components in addition … were included" (p. 13) |
| PA-only interventions included | Y | Y | Y | Y | N | Y | Y | Y |
| OL/Vocab-only interventions included | N | Y | Y | Y | Y | Y | Y | N |
| Fluency-only interventions included | N | Y | Y | Y | Y | Y | Y | N |
| Technology-supported interventions included | N | Y | Y | Y | Y | Y | Y | Y |
| Whole-class interventions included | N | Y | N | N | N | N | Y | Y |
| Dosage/Duration | Any | Minimum duration of 12 weeks | 15–99 sessions | 100 or more sessions | Minimum dosage of 8 hours | Minimum of 15 sessions | Minimum duration of 12 weeks | Minimum of 2 sessions |
| Outcome measure type | Standardized and unstandardized | Standardized and unstandardized[a] | Standardized and unstandardized | Standardized and unstandardized | Standardized and unstandardized[b] | Standardized and unstandardized | Standardized and unstandardized[a] | Standardized only |

*(continued)*

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**TABLE 1**
**Comparing Characteristics of Previous Meta-Analyses of Reading Intervention Research (*continued*)**

| | Suggate (2010) | Slavin et al. (2011) | Wanzek et al. (2016) | Wanzek et al. (2018) | Gersten et al. (2020) | Donegan and Wanzek (2021) | Neitzel et al. (2022) | Current meta-analysis |
|---|---|---|---|---|---|---|---|---|
| Outcome domains | Prereading (concepts of print, letter naming, PA, pseudoword reading); reading (word reading, reading of connected text, general reading measures); reading comprehension | Combined reading outcomes | Foundational reading skills (PA, phonics, word recognition, fluency); language or reading comprehension[c] | Combined reading outcomes | Word or pseudoword reading; passage reading fluency; reading comprehension | Foundational reading skills (PA, phonics, word reading, spelling, fluency); Comprehension (vocabulary, reading comprehension)[c] | Alphabetics; fluency; reading comprehension; general reading | PA; word or pseudoword reading; spelling; passage reading; reading comprehension |
| PA measures | Y | N | Y | Y | N | Y | N | Y |
| OL/Vocab measures | N | Y | Y | Y | N | Y | N | N |
| Spelling measures | N | N | N | Y | N | Y | Y | Y |
| Moderator variables | Participant characteristics (grade, severity of RDs); intervention characteristics (duration, instructional components, group size, language); outcome measure characteristics (reading domain); research methods (research design, publication year) | None | Participant characteristics (grade); Intervention (dosage, instructional components, group size, implementer) | Participant characteristics (severity of RDs); intervention characteristics (dosage, group size); research methods (study year) | Participant characteristics (grade, severity of RDs); intervention characteristics (instructional components, implementer, scripting, feedback, dosage, group size); outcome measure characteristics (standardized vs. unstandardized, reading domain); research methods (research design, nature of comparison group) | Intervention characteristics (dosage, instructional components, group size, individualization); outcome measure characteristics (standardized vs. unstandardized; research methods (publication year) | Participant characteristics (race/ethnicity, poverty); intervention characteristics (supplemental nature, group size, implementer); outcome measure characteristics (reading domain); research methods (research design, level of assignment) | Participant characteristics (grade); intervention characteristics (dosage, instructional components, multisensory nature, group size); outcome measure characteristics (reading domain); research methods (sample size, study design) |
| Robust variance estimation | N | N | N | Y | Y | N | Y | Y |

*Note.* RD = reading difficulties; PA = phonological awareness; OL = oral language; Y = Yes; N = No. Outcome domains = domains for which effects were reported in disaggregated fashion.
[a]Slavin et al. (2011) and Neitzel et al. (2022) excluded effects on "measures of reading objectives inherent to the experimental program (but unlikely to be emphasized in control groups)" (Slavin et al., p. 5).
[b]In the meta-analysis conducted by Wanzek et al. (2018), only 24 of the 328 effect sizes used were derived from researcher-developed measures.
[c]Studies reported effects in disaggregated fashion on standardized foundational skills measures and unstandardized foundational skills measures, standardized language/reading comprehension measures and unstandardized language/reading comprehension measures.

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reading disabilities or difficulties, or identified with reading disabilities" (p. 1946).

The Suggate (2010) meta-analysis included studies with "samples of disadvantaged students" (p. 1560) in Grades PreK-7, but went further by distinguishing between samples "(a) *at-risk*, as characterized by coming from communities with low socioeconomic status and/or being lower performing readers, or (b) *struggling*, that is, reading at or below the 15th percentile, diagnosed with a reading or learning disability (including dyslexia), or having a discrepancy of at least one standard deviation between intelligence quotient and achievement" (p. 1560) and investigating the degree to which severity of reading risk moderated intervention effects. Similarly, Wanzek et al. (2018) included studies with participant samples of Grade K-3 "students with learning disabilities or students identified as at risk for RDs (e.g., students with low ability, low phonemic awareness, language disorders)" (p. 615), but attempted to investigate the degree to which sample mean pretest performance on standardized measures predicted intervention effects.

Gersten et al. (2020) had the most stringent participant risk criterion for inclusion, one that was at least partially norm-referenced: They required included studies to have Grade 1–3 students with "a score on a valid screener or screening battery indicating that the student was likely to be at risk for reading failure at the end of the school year" or "a score on a norm-referenced standardized test … indicating that the student performed below the 40th percentile at the beginning of the school year or at the end of the previous school year" (p. 406). Gersten et al. (2020) further coded studies according to participants' reading "risk level," defined as a two-level variable such that performance at the 25th percentile on a norm-referenced screener distinguished between "at-risk" and "minimal risk" samples (p. 409).

Moderator analyses that investigated the effects of severity of RDs on intervention effects tended to exclude a large number of studies that did not provide enough information to estimate participants' initial RDs. For example, Gersten et al. reported that only 16 included studies provided the information necessary for coding participants' risk level according to their criteria: 12 included studies had samples that were at more risk for RDs and four had samples that were at less severe risk; the remaining 17 studies did not provide enough information to assess the level of participating students' risk. In the Wanzek et al.'s (2018) meta-analysis, only 12 of the 25 included studies reported sample pretest scores on standardized measures such that they could be used to analyze the association between pretest performance and intervention effects. Perhaps most importantly, neither Suggate (2010), Wanzek et al. (2018), nor Gersten et al. (2020) differentiated between initial performance on screeners that measured language or reading comprehension and those that measured word reading, spelling, or skills foundational to word reading or spelling.

### Interventions

As Table 1 illustrates, the seven meta-analyses also differed in terms of acceptable intervention content domains, dosage, settings, and group size. For example, some teams excluded studies of interventions that only addressed PA (Gersten et al., 2020) or only addressed oral reading fluency (Suggate, 2010). Only Slavin et al. (2011) and Neitzel et al. (2022) examined the effects of whole-class interventions.

### Outcomes

Finally, the seven reviews differed in terms of reading outcome domains included in analyses and outcome domains for which disaggregated effects were reported. Only four reviews (Donegan & Wanzek, 2021; Suggate, 2010; Wanzek et al., 2016, 2018) included effects on measures of PA and oral language/vocabulary in their analyses. Only Donegan and Wanzek (2021), Wanzek et al. (2018), and Neitzel et al. (2022) included effects on spelling outcomes in their estimates. When it came to reporting mean effect sizes, most authors (except for Slavin et al., 2011 and Wanzek et al., 2018) reported effects for separate reading domains. At the same time, each author team defined these reading domains differently. For example, Donegan and Wanzek (2021) and Wanzek et al. (2016) reported effects on "foundational skills" and "language/comprehension" (pp. 559–560) outcomes; Gersten et al. (2020) reported effects on "word or pseudoword reading," "passage reading fluency," and "reading comprehension" (p. 415) outcomes. All seven meta-analyses included effects on both researcher-developed/non-standardized measures and standardized, norm-referenced measures in their analyses; Wanzek et al. (2016) were the only team that reported disaggregated effects for each measure type.

## Study Purpose and Research Questions

The most important way in which the present meta-analysis differed from the seven previous meta-analyses described above is our narrowed inclusion criteria regarding the participant sample: We required that students have or be at risk for word RDs as measured by a norm-referenced measure of word reading, spelling, or skills foundational to word reading or spelling. In this way, we hoped to estimate intervention effects for students with or at risk for dyslexia specifically, rather than students with RDs broadly. In addition, unlike previous meta-analyses described above and in Table 1, we only reported the effects on standardized, norm-referenced measures (i.e., we excluded studies that only employed researcher-developed



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measures). We made the decision to report effects on norm-referenced measures because such measures are less likely than researcher-developed measures to be highly aligned to treatment. Although researcher-developed measures that are closely aligned to content taught during an intervention can provide important insight when evaluating intervention effectiveness (Clemens & Fuchs, 2022), such measures can also vary widely across studies. Using norm-referenced measures enabled us to more confidently compare (and calculate mean intervention effects across) studies and participants. We included effects on spelling outcomes in our estimates, as spelling difficulties are endemic to students with or at risk for dyslexia and are frequently targeted in interventions.

We required studies to have a foundational skills component alone or in addition to other intervention components, because we were interested in measuring the effects of interventions that directly addressed the word reading or spelling difficulties experienced by students with dyslexia. Thus, like Suggate (2010), we excluded interventions that only addressed passage reading fluency or only addressed vocabulary/oral language. Unlike Suggate, we also excluded studies that only addressed reading comprehension (i.e., without also addressing word reading, spelling, or skills foundational to word reading or spelling).

Like previous meta-analyses, we explored the potential moderating effects of participant characteristics (i.e., grade level), intervention characteristics (i.e., instructional components, dosage, group size), outcome measure characteristics (i.e., reading domain), and research methods (i.e., sample size, research design). We also explored the effects of moderators that have not been examined previously. In addition to asking whether intervention effects differed when spelling was included as an instructional component, we asked about the effects of instruction on outcomes for students with or at risk for dyslexia when morphological awareness or vocabulary instruction were instructional components. Both cross-sectional (e.g., Tunmer & Chapman, 2012) and longitudinal (e.g., Kendeou et al., 2009) research suggests that oral vocabulary knowledge contributes to word reading. Morphological knowledge has also been found to contribute to word reading, even when controlling for the influence of phonological processing and orthographic knowledge (Deacon, 2012). When students read an unfamiliar word, partial decoding will often be close enough to the correct phonological form that they will be able to arrive at a correct identification if the word is in their listening vocabulary. Similarly, morphological knowledge can help students segment an unfamiliar word into decodable chunks. As Perfetti (2007) articulates in his Lexical Quality Hypothesis, the accuracy and fluency of a student's word recognition depends on the quality of the word's semantic, orthographic, and phonological representation in a student's mental lexicon.

Finally, we were interested in determining whether the effects of interventions that describe themselves as "multisensory" differ from the effects of interventions that do not. Recent legislative mandates and policy briefs that call for teacher training and provision of evidence-based instruction for students with dyslexia explicitly support the use of multisensory instruction. Stevens et al. (2020) reported that at least 20 states have passed legislation explicitly calling for training in and provision of instruction for students with dyslexia that is multisensory in nature. In addition, parents of students with dyslexia often request multisensory reading interventions and litigation resulting from parents' unmet requests for multisensory instruction has increased in recent decades (Rose & Zirkel, 2007).

The purpose of our study, then, was to conduct a meta-analysis that would help to estimate the mean effect of interventions and identify evidence-based intervention components for elementary-grade students with or at risk for dyslexia. We asked:

1. What is the mean effect of reading interventions that include instruction in foundational skills on reading outcomes?

2. Are intervention effects moderated by participant characteristics (i.e., grade level), intervention characteristics (i.e., dosage, content foci, multisensory elements, group size), outcome measure characteristics (i.e., reading domain), or research methods (i.e., sample size, research design)?

# Method

## Search

To identify studies for inclusion, we searched peer-reviewed research articles and grey literature published in English on or after January 1, 1980 and before January 1, 2021, following PRIMSA guidelines (Liberatti et al., 2009). We had two primary reasons for including studies published since 1980. First, we wanted to err on the side of including older research. We examined previous reading intervention research reviews and found that the earliest start date for searches was typically 1980 (e.g., Ritchey & Goeke, 2006; Scammacca et al., 2015). Second, the Education for All Handicapped Children Act was signed into law in 1975, which helped to kickstart research for children with disabilities in the 1980s.

The search included a three-step process. First, we searched the electronic databases of ERIC and PsycINFO with a combination of the following Boolean search terms, specifying that abstracts must include one of the following population identifiers (*dyslex**, *reading difficult**, *reading disabilit**, *risk for reading difficult**, *risk for reading disabilit**, *reading delay**, *reading disorder**, *learning disab**) and a term that described reading interventions (*reading*



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*interven*, reading instruction, phon*, correspondence*, reading fluency*). See Appendix A for a description of databases and search terms used in the previous meta-analyses (see Table 1) to which we draw comparisons. This search yielded 13,752 unique articles. After a 1-hour training, nine screeners used the Covidence systematic review software (Covidence, 2021) to screen abstracts. If information needed to ascertain eligibility for inclusion was not provided in the article abstract, studies were recommended for full-text review. All abstracts were independently screened by at least two screeners; the first author reviewed and resolved screening disagreements.

After the conclusion of abstract screening, 541 full texts were retrieved for review. Before beginning full-text review, seven reviewers participated in a 3-hour training and achieved >90% reliability with the first author when reviewing a practice set of three articles. Reviewers were expected to apply inclusion criteria in a pre-specified order and identify the same reason for exclusion when exclusion was appropriate. All texts were independently reviewed by two reviewers and disagreements were resolved by the first author.

Studies were included if they met the following criteria:

1. Publication date was on or after January 1, 1980 and before January 1, 2021.
2. Source articles used screening criteria requiring that participants perform at or below the 25th percentile on a norm-referenced test of word reading, spelling, or foundational skills (i.e., PA, knowledge of grapheme–phoneme correspondences, real word reading, pseudoword reading, spelling, or passage reading accuracy or fluency). Source articles were also included when the meta-analysis team determined that the study sample had a mean baseline score at or below the 16th percentile on at least one of the norm-referenced test types listed above. Studies including students without foundational skill difficulties were included when disaggregated data were provided for the subset of students with these difficulties. For these studies, only disaggregated data were used to calculate effect sizes. Studies were excluded if participants had intellectual disabilities, autism spectrum disorders, or sensory disabilities (e.g., deafness, blindness).
3. Participants were in Grades K-5 (ages 5–11), or the sample mean age was within this range.
4. Studies employed experimental or quasi-experimental, treatment-comparison or multiple treatment-comparison research designs with at least 15 participants per group. For quasi-experimental design studies to be included, we required that groups be equivalent at pretest (i.e., there could be no statistically significant difference between groups at baseline on a reading measure). Acceptable comparison conditions included (a) "business as usual" classroom reading instruction, (b) "business as usual" exposure to non-reading instruction or study hall, (c) a researcher-manipulated non-reading treatment (e.g., math, attention-training), or (d) a researcher-manipulated reading treatment that did not involve instruction in PA, word reading, or spelling. Comparison conditions that consisted of different dosages of the "treatment" intervention and historical comparison conditions were excluded.
5. Instruction took place across more than one session and focused on PA, phonics, word reading, spelling, or a combination of these domains. Studies that had other, additional intervention components were included.
6. The primary language of instruction was English (i.e., greater than 50% of instruction was delivered in English). Studies with interventions that incorporated first language supports within English-language instruction were included.
7. The intervention setting was school based (i.e., this included school-based after-school tutoring and school-based summer instruction). Parent-delivered reading interventions were not included, nor were interventions delivered in clinical settings.
8. The study had at least one calculable effect size on a norm-referenced measure of PA, knowledge of grapheme–phoneme correspondences, pseudoword reading, real word reading, spelling, or passage reading.

Of the 541 studies reviewed, 38 met all inclusion criteria. Studies were excluded because they did not meet at least one eligibility criterion, applied in the following order: enrolled students with mean age outside of the eligible age range ($k$ = 35); implemented fewer than two intervention sessions ($k$ = 2); did not address PA, phonics, word reading, or word spelling ($k$ = 29); were conducted in a language other than English ($k$ = 35); were not conducted in a school setting ($k$ = 26); did not employ an eligible, norm-referenced outcome measure ($k$ = 24); employed an ineligible study design ($k$ = 158); did not provide enough information to calculate effect size ($k$ = 32); did not enroll at least 15 participants per group ($k$ = 12); enrolled students who did not have word RDs ($k$ = 105); reported the results of analyses already reported in another publication ($k$ = 43); and reported follow-up outcomes only (i.e., for studies that had immediate intervention effects reported in a separate publication; $k$ = 2). The order in which full-text review criteria were applied impacted these numbers (e.g., any of the 32 studies excluded because they did not provide enough information to calculate effect size may have also failed to meet the subsequent requirements). We



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did not attempt to contact authors of studies to retrieve information that was not reported in articles (e.g., statistics needed to calculate effect sizes or determine whether student participants met the RDs criterion).

After the initial search, we completed a hand search of the five journals that published the greatest number of included studies (*Exceptional Children*, *Journal of Educational Psychology*, *Journal of Learning Disabilities*, *Journal of Research on Educational Effectiveness*, and *Learning Disabilities Research & Practice*). This hand search included articles published on or after January 1, 2016 and before January 1, 2021. Based on the hand search, two additional articles were identified for review; both were ultimately excluded.

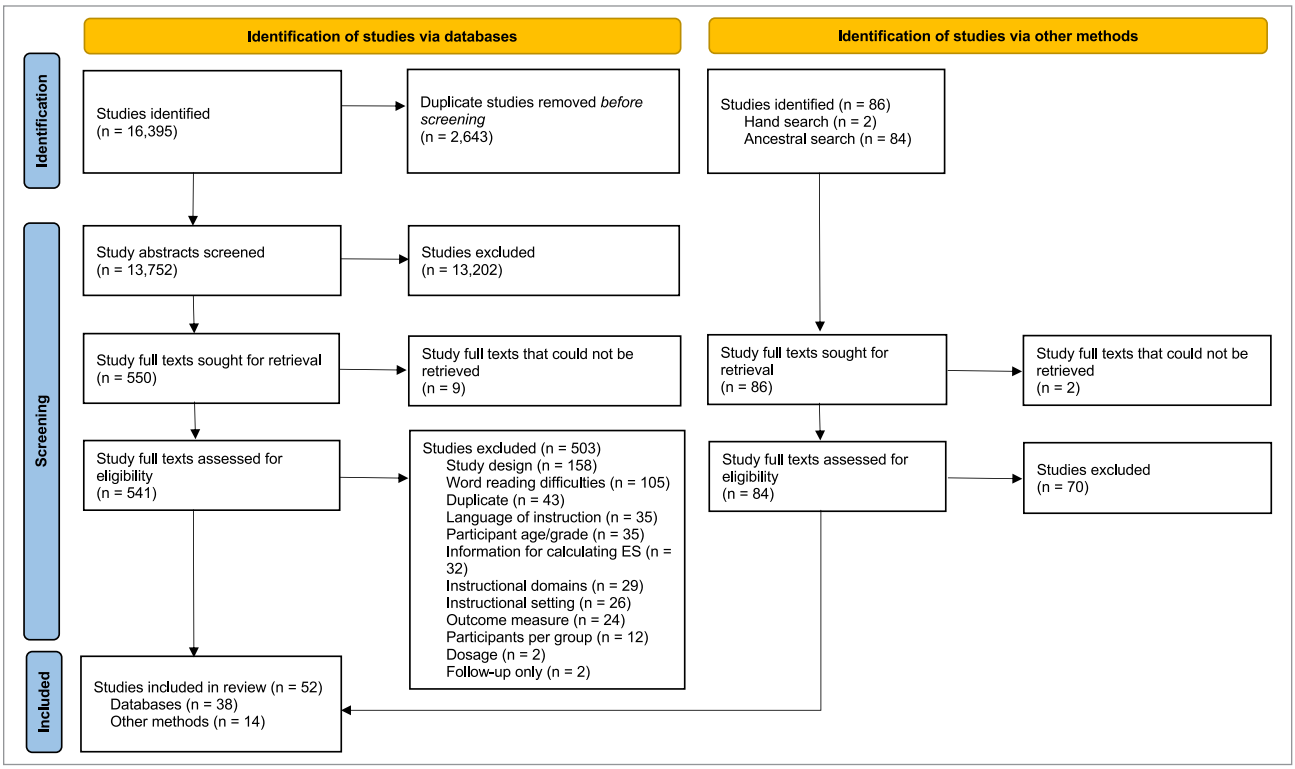
Finally, we completed an ancestral search of articles included in relevant literature reviews conducted in the last 5 years (Gersten et al., 2020; Suggate, 2016; Wanzek et al., 2016, 2018). After an initial review of abstracts, the team reviewed the full text of 82 articles; 14 additional peer-reviewed articles met inclusion criteria and were coded. Thus, a total of 53 studies within 52 publications (38 identified through the database search and 14 identified through the ancestral search) were identified for inclusion. Figure 1 represents our search procedure and results at each stage of the search process.

## Coding Procedures

We coded for characteristics of participants, interventions, outcome measures, and research methods within included studies. Information about definitions used to code for these variables is provided in Appendix B (see supplemental online materials). Eight coders participated in a 3-hour training and independently coded articles until they obtained a minimum of 90% reliability with the first author in each code sheet section. Once coding began, all articles were independently double coded by two members of the author team. When coding discrepancies arose, the first author resolved disagreements.

Coding procedures for the multisensory instruction variable may merit further explanation. To be coded as a "multisensory" intervention, interventions needed to be explicitly described by the author team as using a "multisensory" approach. We coded for the variable in this way because there is a lack of consensus as to what constitutes multisensory instruction (Fletcher, 2018), and districts acting on previously described legislative mandates requiring that students with dyslexia be provided with multisensory instruction are likely to select a program that explicitly describes itself as multisensory.

**FIGURE 1**
**PRISMA Search Flow Diagram**






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## *Effect Size Calculation and Meta-Analytic Procedures*

To quantify the effects of interventions on reading outcomes for students with or at risk for dyslexia in Grades K-5, we used standardized mean differences between intervention and control groups estimated with Hedges' *g* (Hedges, 1981), using study-reported posttest mean (M) and standard deviation (SD) estimates for each condition. We used adjusted posttest means (i.e., posttest means adjusted for pretest scores) when they were available. When only test statistics (*F* statistics) were provided, *F* values were transformed to Hedges' *g* effect sizes. When calculating study-level effect sizes, if there was more than one measure within a given outcome domain (i.e., more than one measure of PA, decoding/encoding, connected-text reading, or reading comprehension), we calculated a mean effect size using a random effects model that employed a restricted maximum likelihood approach to estimate variance. All analyses were conducted with the *metafor* package (Viechtbauer, 2010) and the *clubSandwich* package (Pustejovsky, 2019) for the *R* statistical computing environment (R Core Team, 2019).

### Outlier Analysis

Prior to estimating average effect sizes across studies, we examined the distribution of raw effect size estimates to identify outliers. We defined an outlier as a value below the first quartile minus three times the interquartile range (−1.24) or above the third quartile plus three times the interquartile range (2.07; Tukey, 1977). We identified one outlier at the upper end of the distribution (2.54; Storey et al., 2020) and winsorized the value to the upper fence value.

### Estimation of Mean Effects

There are several sources of dependencies in our effect size estimates. Many included studies reported intervention effects on multiple measures. Several studies included multiple contrasts because they examined the effects of more than one intervention or control condition. Effect size estimates from a single study are likely to be correlated because they are from the same or a shared sample. Therefore, we used three-level, multivariate random effects analyses, assuming a correlation of .80. In addition, we used robust variance estimation (RVE) to account for dependencies in our data (Hedges et al., 2010; Pustejovsky & Tipton, 2022), applying small-sample corrections to standard errors, hypothesis tests, and confidence intervals (Tipton, 2015; Tipton & Pustejovsky, 2015). In these analyses, we used study as a clustering unit. This analytic approach estimates the average effect size using all information available and then adjusts the standard errors to account for the inherent clustering of these related effect sizes. The model decomposes the variance of the effect size into three parts. The Level 1 variance is the sampling variance estimated via the traditional variance calculation. The Level 2 variance is the within-study variance. The Level 3 variance is the variance between the studies. We examined the possibility of publication bias with a modified version of Egger's regression that accounted for dependent effect size estimates (Egger et al., 1997; Rodgers & Pustejovsky, 2021).

To describe the heterogeneity between studies and within each study, we report the restricted maximum likelihood estimate of the between-study variance ($\tau^2$) and within-study variance ($\omega^2$). We also report the *Q* statistics and $I^2$ statistic partitioned to each level to reveal the extent to which heterogeneity among true effect sizes contributes to the observed variation in the effect size estimates. We adopt the guideline articulated by Higgins et al. (2003), which holds that $I^2$ of 50%–75% indicates a moderate amount of heterogeneity, enough to conduct moderator analyses.

### Moderator Analysis

We first examined each moderator separately using meta-regression, as this single moderator approach has predominated in previous meta-analyses (e.g., Gersten et al., 2020; Wanzek et al., 2016) and single moderator analyses can provide meaningful point estimates for each level of the categorical moderators. The effects of moderators were tested using small-sample-adjusted *t*-tests for the moderators with two categories. For the moderators with more than two categories, we used the Wald test function from the *clubSandwich* package (Pustejovsky, 2019) that applies small-sample-adjusted *F* tests for the moderators with more than two categories. Next, we conducted a multiple meta-regression analysis to model the effects of the moderators simultaneously. This approach allowed us to look at the impact of each moderator while controlling for other potential moderators; it provided us with information as to how moderator effects should be interpreted in the context of other moderators.

# Results

Table 2 presents participant and study characteristics for the 53 studies included in this meta-analysis, which encompassed 351 individual effect sizes and a combined sample of 6,053 students (M = 114.21, SD = 79.45, range = 32–422). We provide intervention characteristics alongside of study-level effect sizes in Table 3.

At least partly as a result of our search years, decision to include grey literature, and differences in inclusion criteria compared with those articulated in previous meta-analyses (see Table 1), only 14 of the 53 studies we identified for



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**TABLE 2**
**Participant and Study Characteristics**

| Study | *N* | Grade | Design | Fidelity |
|---|---|---|---|---|
| Al Otaiba et al. (2005) | 73 | K | RCT | Y |
| Baker et al. (2000) | 84 | 1–2 | RCT | N |
| Blachman et al. (2004) | 69 | 2–3 | RCT | Y |
| Burns (2011) | 78 | 1 | RCT | Y |
| Christodoulou et al. (2017) | 47 | 1–4 | RCT | N |
| Coyne et al. (2013) | 162 | K | CRT | Y |
| Denton et al. (2010) | 422 | 1 | RCT | Y |
| Denton et al. (2014) | 206 | 1 | RCT | Y |
| Donegan et al. (2020) Study 1 | 153 | 4 | RCT | Y |
| Duff et al. (2014) | 145 | 1 | QED | Y |
| Fawcett et al. (2001) | 87 | 2 | QED | N |
| Fives et al. (2013) | 227 | 2 | RCT | Y |
| Frantz (2000) | 78 | 2–6 | QED | N |
| Georgiou et al. (2020) | 48 | 3 | RCT | Y |
| Graham et al. (2002) | 54 | 2 | RCT | Y |
| Gunn et al. (2005) | 245 | K-3 | RCT | Y |
| Hagans and Good (2013) | 50 | 1 | RCT | N |
| Hatcher et al. (2006) | 77 | K | RCT | N |
| Jenkins et al. (2004) | 99 | 1 | RCT | Y |
| Little et al. (2012) | 90 | K | RCT | Y |
| Lovett et al. (2017) | 219 | 1–3 | QED | Y |
| Mathes et al. (2003) | 89 | 1 | QED | Y |
| Mayfield (2000) | 60 | 1 | RCT | N |
| Miciak et al. (2018) | 270 | 4 | RCT | Y |
| Morris et al. (2012) | 279 | 2–3 | RCT | N |
| Nicolson et al. (1999) | 102 | 1 | QED | N |
| O'Callaghan et al. (2016) | 98 | PK-K | RCT | N |
| O'Connor et al. (2002) | 46 | 3–5 | RCT | Y |
| O'Shaughnessy and Lee Swanson (2000) | 45 | 2 | RCT | Y |
| Scanlon et al. (2005) | 84 | K-1 | RCT | N |
| Simmons et al. (2011) | 206 | K | CRT | Y |
| Storey et al. (2020) | 32 | 1–4 | RCT | N |
| Torgesen et al. (2010) | 108 | 1 | RCT | N |
| Toste et al. (2019) | 108 | 4–5 | RCT | Y |
| Vadasy, Jenkins, Antil, and Wayne (1997) | 35 | 1 | RCT | N |
| Vadasy, Jenkins, Antil, Wayne, and O'Connor (1997) | 40 | 1 | RCT | Y |
| Vadasy et al. (2000) | 46 | 1 | RCT | Y |
| Vadasy et al. (2005) | 57 | 1 | QED | Y |
| Vadasy et al. (2006) | 67 | K | RCT | Y |
| Vadasy et al. (2007) | 43 | 2–3 | RCT | Y |
| Vadasy and Sanders (2008a) | 86 | K | QED | Y |
| Vadasy and Sanders (2008b) | 162 | 2–3 | RCT | Y |
| Vadasy and Sanders (2009) | 202 | 2–3 | RCT | Y |
| Vadasy and Sanders (2010) | 84 | K | RCT | Y |
| Vadasy and Sanders (2011) | 98 | 1 | RCT | Y |
| Vaughn, Mathes, et al. (2006) | 40 | 1 | RCT | Y |
| Vaughn, Cirino, et al. (2006) | 91 | 1 | RCT | Y |
| Vellutino et al. (2008) | 113 | K | RCT | N |
| Wang and Algozzine (2008) | 139 | 1 | CRT | Y |
| Wanzek and Vaughn (2008) Study 1 | 50 | 1 | RCT | Y |
| Wanzek and Vaughn (2008) Study 2 | 36 | 1 | RCT | Y |
| Wanzek et al. (2020) | 260 | 4 | RCT | Y |
| Wise et al. (1999) | 153 | 2–5 | QED | N |

*Note.* RCT = randomized controlled trial; CRT = cluster randomized trial; QED = quasi-experimental design; Y = Yes; N = No.

inclusion were included by Suggate (2010), 13 of the 53 were included by Slavin et al. (2011), 22 of the 53 were included by Wanzek et al. (2016), 10 of the 53 were included by Wanzek et al. (2018), 12 of the 53 were included by Gersten et al. (2020), and 15 of the 53 were included by Neitzel et al. (2022). A representation of the overlap (or lack thereof) between articles included in the seven meta-analyses is represented in Appendix C (see supplemental online materials). Despite the fact that January 1, 1980 was the start date for our search, all articles that met inclusion



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**TABLE 3**
**Intervention Characteristics**

| Study | Treatment name | Instructional components | Implementer | Group size | Dosage | PA ES | D/E ES | TR ES | RC ES |
|---|---|---|---|---|---|---|---|---|---|
| Al Otaiba et al. (2005) | Tutor Assisted Intensive Learning Strategies | PA, D, RR, TR, RC, V | O | 1:1 | 64 | 0.46 | 0.53 | | 0.71 |
| Al Otaiba et al. (2005) | Tutor Assisted Intensive Learning Strategies | PA, D, RR, TR, RC, V | O | 1:1 | 32 | 0.57 | 0.22 | | 0.37 |
| Baker et al. (2000) | Start Making a Reader Today | D, TR, RC | O | 1:1 | 24 | | 0.40 | | 0.28 |
| Blachman et al. (2004) | Intensive Reading Remediation | D, E, TR, RC, V | R | 1:1 | 133 | | 0.86 | 0.73 | 0.51 |
| Burns (2011) | Systematic and Explicit Teaching Routines | PA, D, RR, E, TR, RC, V | SP | SG | 38 | | −0.12 | −0.10 | 0.02 |
| Christodoulou et al. (2017) | Seeing Stars | PA, D, RR, E, TR, RC | O | SG | 120 | | 0.63 | 0.17 | |
| Coyne et al. (2013) | Early Reading Intervention | PA, D, E, TR | SP | SG | 63 | −0.03 | −0.13 | | |
| Denton et al. (2010) | Responsive Reading Instruction | PA, D, RR, E, TR, RC | SP | SG | 83 | 0.18 | 0.43 | | 0.48 |
| Denton et al. (2014) | Guided Reading | D, RR, TR, RC | R | SG | 72 | | 0.23 | | 0.06 |
| Denton et al. (2014) | Explicit Instruction | D, RR, M, E, TR, RC, V | R | SG | 72 | | 0.37 | | 0.24 |
| Donegan et al. (2020) Study 1 | Voyager Passport | D, M, E, TR, RC, V | R | SG | 46 | | −0.10 | −0.14 | 0.11 |
| Duff et al. (2014) | Reading and Language Intervention | PA, D, RR, E, TR, RC, V | SP | SG | | −0.11 | 0.13 | −0.11 | 0.10 |
| Fawcett et al. (2001) | Interactive Assessment and Teaching | PA, D, E, TR, RC, V | R | SG | | | 0.58 | | |
| Fives et al. (2013) | Wizard of Words | PA, D, TR, RC, V | O | 1:1 | 29 | | 0.09 | | |
| Frantz (2000) | Auditory Discrimination in Depth | PA, D, E | R | SG | 20 | | −0.07 | | 0.53 |
| Frantz (2000) | Reciprocal Teaching | PA, D, RC | R | SG | 20 | | 0.42 | | −0.01 |
| Georgiou et al. (2020) | Structured Word Inquiry | D, M, E, V | R | 1:1 | 15 | | 0.42 | | |
| Georgiou et al. (2020) | Simplicity | PA, D, M, E, TR, V | R | 1:1 | 15 | | 0.13 | | |
| Graham et al. (2002) | Spelling Instruction | D, M, E | R | SG | 16 | | 0.66 | | |
| Gunn et al. (2005) | Reading Mastery or Corrective Reading | PA, D, E, TR | R | SG | 145 | | 0.65 | | 0.57 |
| Gunn et al. (2005) | Reading Mastery or Corrective Reading | PA, D, E, TR, V | R | SG | 145 | | 0.20 | | −0.07 |
| Hagans and Good (2013) | Early Literacy Intervention | PA, D | R | SG | 15 | 1.11 | −0.14 | 0.25 | |
| Hatcher et al. (2006) | Sound Linkage Reading Intervention | PA, D, RR, TR | SP | SG | 17 | 0.23 | 0.15 | | |

*(continued)*

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**TABLE 3**
**Intervention Characteristics (*continued*)**

| Study | Treatment name | Instructional components | Implementer | Group size | Dosage | PA ES | D/E ES | TR ES | RC ES |
|---|---|---|---|---|---|---|---|---|---|
| Jenkins et al. (2004) | Sound Partners | D, RR, E, TR | SP | 1:1 | 50 | | 0.58 | | 0.85 |
| Jenkins et al. (2004) | Sound Partners | D, RR, E, TR | SP | 1:1 | 50 | | 0.55 | | 0.75 |
| Little et al. (2012) | Early Reading Intervention | PA, D, E, TR | SP | SG | 80 | 0.36 | 0.22 | | |
| Lovett et al. (2017) | Triple-Focus | PA, D, M, E, TR, RC, V | R | SG | 160 | | 0.96 | 0.79 | 0.88 |
| Mathes et al. (2003) | Peer-Assisted Learning Strategies | PA, D, RR, TR, RC | SP | SG | 28 | 1.57 | 0.59 | | 0.42 |
| Mathes et al. (2003) | Peer-Assisted Learning Strategies | PA, D, RR, TR, RC | SP | SG[a] | 24 | 1.61 | 0.73 | | 0.70 |
| Mayfield (2000) | Edmark Reading Program | RR, M, TR, RC, V | O | 1:1 | 20 | | 0.07 | | 0.56 |
| Miciak et al. (2018) | Intensive Reading Intervention | D, RR, E, TR, RC, V | R | SG | 47 | | −0.11 | | −0.21 |
| Morris et al. (2012) | Phonological and Strategy Training | PA, D, M | R | SG | 70 | | 0.59 | 0.19 | 0.51 |
| Morris et al. (2012) | Phonological Analysis and Blending/Direct Instruction + Retrieval, Automaticity, Vocabulary, Engagement with Language, and Orthography | PA, D, M, E, TR, RC, V | R | SG | 70 | | 0.65 | 0.59 | 0.46 |
| Morris et al. (2012) | Phonological Analysis and Blending/Direct Instruction + Classroom Survival Skills | PA, D | R | SG | 70 | | 0.22 | 0.20 | 0.25 |
| Nicolson et al. (1999) | Interactive Assessment and Teaching | PA, D, RR, E, TR, RC, V | CD | SG | 10 | | 0.57 | | |
| O'Callaghan et al. (2016) | Lexia Core5 | PA, D, RR, E, TR, RC, V | CD | 1:1 | 17 | 0.37 | 0.33 | | |
| O'Connor et al. (2002) | Reading-Level Matched | PA, D, E, TR, RC | SP | 1:1 | 36 | | 1.18 | | 1.45 |
| O'Connor et al. (2002) | Grade-Level Matched | PA, D, E, TR, RC, V | SP | 1:1 | 36 | | 1.06 | | 1.21 |
| O'Shaughnessy and Lee Swanson (2000) | Phonological Awareness Training | PA, D, E | SP | SG | 9 | 1.86 | 0.47 | | 0.65 |
| O'Shaughnessy and Lee Swanson (2000) | Word Analogy Training | PA, D, E, TR | SP | SG | 9 | 1.19 | 0.38 | | 0.74 |
| Scanlon et al. (2005) | Text Emphasis | PA, D, RR, E, TR | SP | 1:1 | 160 | | 0.60 | | 0.50 |
| Scanlon et al. (2005) | Phonological Skills Emphasis | PA, D, RR, E, TR | SP | 1:1 | 160 | | 0.68 | | 0.17 |
| Simmons et al. (2011) | Early Reading Intervention | PA, D, E, TR | SP | SG | 63 | 0.30 | 0.19 | | −0.12 |
| Storey et al. (2020) | Headsprout Early Reading | PA, D, TR, RC, V | CD | 1:1 | 70 | | 2.54 | 0.91 | |

*(continued)*

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**TABLE 3**
**Intervention Characteristics (*continued*)**

| Study | Treatment name | Instructional components | Implementer | Group size | Dosage | PA ES | D/E ES | TR ES | RC ES |
|---|---|---|---|---|---|---|---|---|---|
| Torgesen et al. (2010) | Lindamood Phoneme Sequencing Program for Reading, Spelling, and Speech | PA, D, E, TR | CD | SG | 107 | 0.67 | 0.80 | | 0.53 |
| Torgesen et al. (2010) | Read, Write, and Type | PA, D, E, TR | CD | SG | 107 | 0.56 | 0.42 | | 0.38 |
| Toste et al. (2019) | Multisyllabic Word Reading | D, RR, M, E, TR, V | R | SG | 27 | | 0.28 | | 0.06 |
| Toste et al. (2019) | Multisyllabic Word Reading Intervention with Motivational Belief Training | D, RR, M, E, TR, V | R | SG | 27 | | 0.58 | | 0.15 |
| Vadasy, Jenkins, Antil, and Wayne (1997) | Sound Partners | PA, D, E, TR | O | 1:1 | 54 | | 0.56 | | |
| Vadasy, Jenkins, Antil, Wayne, and O'Connor (1997) | Sound Partners | PA, D, RR, M, E, TR | O | 1:1 | 46 | | 0.25 | | |
| Vadasy et al. (2000) | Sound Partners | PA, D, E, TR, RC | O | 1:1 | 54 | | 0.94 | | |
| Vadasy et al. (2005) | Sound Partners | PA, D, RR, M, E, TR | SP | 1:1 | 64 | | 0.80 | | 0.82 |
| Vadasy et al. (2005) | Sound Partners | PA, D, RR, M, E | SP | 1:1 | 64 | | 0.83 | | 0.58 |
| Vadasy et al. (2006) | Code-Oriented Instruction | PA, D, RR, E, TR | SP | 1:1 | 36 | 0.26 | 0.65 | | 0.27 |
| Vadasy et al. (2007) | QuickReads | D, RR, E, TR, RC | SP | 1:1 | 30 | | 0.34 | 0.51 | |
| Vadasy and Sanders (2008a) | Code-Oriented Instruction | PA, D, RR, E, TR | SP | 1:1 | 36 | 0.37 | 0.33 | | 0.39 |
| Vadasy and Sanders (2008a) | Code-Oriented Instruction | PA, D, RR, E, TR | SP | SG | 36 | 0.63 | 0.55 | | 0.46 |
| Vadasy and Sanders (2008b) | QuickReads | D, RR, TR, RC | SP | SG | 30 | | −0.03 | 0.18 | −0.01 |
| Vadasy and Sanders (2009) | QuickReads | D, RR, TR, RC, V | SP | SG | 30 | | 0.14 | 0.28 | 0.19 |
| Vadasy and Sanders (2009) | QuickReads | D, RR, TR, RC, V | SP | SG | 30 | | −0.11 | 0.17 | 0.18 |
| Vadasy and Sanders (2010) | Phonics-Based Instruction | PA, D, RR, E, TR, V | SP | 1:1 | 36 | 0.03 | 0.48 | | 0.47 |
| Vadasy and Sanders (2011) | Sound Partners | PA, D, RR, E, TR, V | SP | 1:1 | 40 | 0.18 | 0.20 | | 0.10 |

*(continued)*

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**TABLE 3**
**Intervention Characteristics (*continued*)**

| Study | Treatment name | Instructional components | Implementer | Group size | Dosage | PA ES | D/E ES | TR ES | RC ES |
|---|---|---|---|---|---|---|---|---|---|
| Vaughn, Mathes, et al. (2006) | Proactive Reading | PA, D, RR, E, TR, RC, V | SP | SG | 133 | 0.82 | 0.90 | 0.17 | 1.06 |
| Vaughn, Cirino, et al. (2006) | Proactive Reading | PA, D, RR, TR, RC, V | SP | SG | 133 | 0.19 | 0.31 | 0.29 | 0.06 |
| Vellutino et al. (2008) | Project-Based Intervention | PA, D, RR, E, TR | SP | SG | 28 | 0.95 | 0.50 | | |
| Wang and Algozzine (2008) | Targeted Intervention | PA, D, RR, TR | SP | SG | 33 | 0.46 | 0.26 | | 0.35 |
| Wanzek and Vaughn (2008) Study 1 | Reading Intervention | D, E, TR, RC, V | R | SG | 33 | | 0.06 | −0.21 | −0.07 |
| Wanzek and Vaughn (2008) Study 2 | Reading Intervention | D, E, TR, RC, V | R | SG | 65 | | 0.13 | −0.05 | −0.63 |
| Wanzek et al. (2020) | Voyager Passport | D, RR, E, TR, RC, V | R | SG | 75 | | 0.22 | 0.18 | 0.08 |
| Wise et al. (1999) | Phonological Awareness Training | PA, D, E, TR | CD | SG | 60 | | 0.43 | | |

*Note.* PA = phonological awareness; D = decoding; RR = rote reading of whole words; M = morphology; E = encoding; TR = connected-text reading fluency; RC = reading comprehension; V = vocabulary; O = other implementer (e.g., community volunteer); R = researcher-delivered; SP = school personnel; CD = computer-delivered; 1:1 = one-on-one; SG = small-group; ES = effect size (all effect sizes were calculated using the formula for Hedges' *g*).
[a]This study was the only study for which the teacher-to-student ratio during the intervention session differed from the group size.

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criteria were published after 1996. As others have noted (Gersten et al., 2005), research quality standards were still emerging during the 1980s and 1990s. Because our search required that studies employ norm-referenced measures and experimental or quasi-experimental designs with at least 15 participants per group, studies from these decades were more likely to be excluded.

In all, 12 studies (23%) included students in kindergarten, 27 (51%) included first graders, 16 (30%) included second graders, 13 (25%) included third graders, nine (17%) included fourth graders, and four (8%) included fifth graders. In total, 16 of the studies (30%) included participants in two or more grades. In 12 of the studies (23%), 50% or more of the participants experienced economic disadvantage (measured as percentage of participating students eligible to receive free or reduced-price school lunch) and in five of the studies (9%), fewer than 50% of participants experienced economic disadvantage; in 36 studies (68%), no information about the socioeconomic status of the participant sample was provided. Five studies (9%) included at least 50% of students who were emergent multilingual students learning English (ELs); for the remaining studies, fewer than 50% of students were ELs (27 studies; 50%) or not enough information was reported to determine the number of ELs in the sample (19 studies; 36%).

All qualifying studies had designs that compared treatment groups to comparison or control groups; 41 (77%) were randomized control trials (RCTs), three (6%) were cluster-randomized trials, and nine (17%) were quasi-experimental research designs (QEDs). In 37 of the studies (70%), researchers described the collection of data on fidelity of intervention implementation and reported the results in the paper.

The 53 included studies provided effect sizes associated with 70 intervention treatment conditions. Table 3 describes intervention features for the 70 contrasts. The interventions examined most frequently included Sound Partners (eight treatment conditions) and QuickReads (four treatment conditions). Most (63%) interventions examined were delivered to small groups of students, with a mean small group size of 3.55 (SD = 1.11). The remaining contrasts (37%) examined the effects of one-on-one interventions. Interventions were implemented by school personnel in 46% of included contrasts; 33% were delivered by researchers or researcher-hired interventionists; 9% were computer-delivered or technology-supported; and 13% were delivered by someone who did not fall into these categories (e.g., a community volunteer). Across studies, intervention was provided for an average of 55.19 hours (SD = 40.75, range = 9–160 hours).

Although our inclusion criteria did not require this to be the case, all 70 interventions included word reading instruction. One study (Mayfield, 2000) only taught students to read words by rote. All other studies provided word reading instruction that addressed knowledge of grapheme–phoneme correspondences and decoding. Some interventions also included a PA instruction component (71%); 20% included instruction in morphology, 73% in spelling, 87% in connected-text reading, 50% in reading comprehension, and 41% in vocabulary. Five of the 70 interventions (7%) were described as using a "multisensory" approach.

# Meta-Analytic Findings

## Main Effects Analyses

The weighted average effect on combined outcomes was estimated as $g = 0.33$ (95% CI [0.25, 0.41], $p < .01$). As expected, there was substantial heterogeneity across effect sizes ($Q = 1747.03$, $df = 350$, $p < .01$) with between-study variance of $\tau^2 = 0.03$ and within-study variance of $\omega^2 = .05$, and a total $I^2$ of 63% (between-study = 23% and within-study = 40%).

## Single Moderator Analyses

We report estimates of average effect size disaggregated by the levels of each moderator in Table 4.

### *Student Characteristics*

The mean effect of intervention did not statistically significantly differ based on student grade level ($t = 1.82$, $df = 7.55$, $p = .11$). Descriptively speaking, studies with students in Grades 3–5 tended to have smaller effects ($g = 0.16$) than studies with students in Grades K-2 ($g = 0.36$); however, this difference was not statistically significant.

### *Intervention Characteristics*

Of the instructional characteristics explored, only dosage and inclusion of a spelling instructional component were statistically significant moderators of effect sizes. The effects of dosage ($t = 2.21$, $df = 12.40$, $p < .05$) indicated that, with each additional hour of intervention, effect size tended to increase by 0.002. Interventions that provided spelling instruction in addition to word reading instruction had larger effects ($g = 0.37$) than those without a spelling instruction component ($g = 0.23$; $t = 2.93$, $df = 5.03$; $p = .03$).

Inclusion of a PA component in addition to a word reading component was not statistically significantly more effective than providing word reading instruction alone; however, descriptively speaking, interventions with a PA component outperformed ($g = 0.38$) those without a PA component ($g = 0.22$; $t = 2.03$, $df = 25.15$, $p = .05$). Interventions that provided morphology or vocabulary instruction as well as providing word reading instruction ($g = 0.34$) yielded very similar effects to word reading interventions without either component ($g = 0.33$; $t = 0.08$; $df = 8.18$, $p = .93$). The effects of interventions that were explicitly described as multisensory ($g = 0.20$) did not differ from



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**TABLE 4**
**Results of the Separate, Single Moderator Analyses**

| Moderator | *m* | *k* | Effect size (beta) | SE | *df* | *p* | 95% CI | *t/F* | *df* | *p* |
|---|---|---|---|---|---|---|---|---|---|---|
| Participant characteristics | | | | | | | | | | |
| Grade level | | | | | | | | 1.82 | 7.55 | .11 |
| K-2 | 304 | 45 | 0.36 | 0.04 | 38.15 | <.01 | [0.28, 0.44] | | | |
| 3–5 | 47 | 8 | 0.16 | 0.10 | 5.36 | .16 | [−0.09, 0.42] | | | |
| Intervention characteristics | | | | | | | | | | |
| Dosage | | | | | | | | | | |
| Intercept | | | 0.34 | 0.04 | 39.05 | <.01 | [0.26, 0.41] | | | |
| Dosage (hour) | | | 0.002 | 0.001 | 12.40 | .047 | [0.00, 0.004] | | | |
| Group size | | | | | | | | 1.02 | 13.32 | .33 |
| One-on-One | 246 | 34 | 0.30 | 0.05 | 28.77 | <.01 | [0.20, 0.41] | | | |
| Small group | 105 | 20 | 0.40 | 0.07 | 14.57 | <.01 | [0.25, 0.56] | | | |
| Multisensory | | | | | | | | −1.41 | 3.19 | .25 |
| No | 332 | 49 | 0.34 | 0.04 | 41.86 | <.01 | [0.26, 0.42] | | | |
| Yes | 19 | 4 | 0.20 | 0.09 | 2.74 | .14 | [−0.12, 0.51] | | | |
| Morphology/ Vocabulary | | | | | | | | 0.08 | 8.18 | .93 |
| No | 180 | 29 | 0.33 | 0.07 | 25.51 | <.01 | [0.17, 0.48] | | | |
| Yes | 171 | 28 | 0.34 | 0.08 | 24.70 | <.01 | [0.17, 0.50] | | | |
| Spelling | | | | | | | | 2.93 | 5.03 | .03 |
| No | 91 | 15 | 0.23 | 0.04 | 9.32 | <.01 | [0.14, 0.32] | | | |
| Yes | 260 | 41 | 0.37 | 0.05 | 36.59 | <.01 | [0.28, 0.46] | | | |
| PA | | | | | | | | 2.03 | 25.15 | .05 |
| No | 108 | 16 | 0.22 | 0.07 | 12.82 | .01 | [0.14, 0.32] | | | |
| Yes | 243 | 38 | 0.38 | 0.04 | 31.63 | <.01 | [0.28, 0.46] | | | |
| Outcome measure domains | | | | | | | | 4.60 | 3, 21.8 | .01 |
| PA | 42 | 20 | 0.44 | 0.07 | 25.59 | <.01 | [0.29, 0.60] | | | |
| Word reading/ Spelling | 213 | 53 | 0.34 | 0.04 | 45.76 | <.01 | [0.26, 0.42] | | | |
| Text reading | 31 | 17 | 0.25 | 0.05 | 18.39 | <.01 | [0.15, 0.34] | | | |
| Reading comprehension | 65 | 35 | 0.26 | 0.05 | 38.53 | <.01 | [0.16, 0.35] | | | |
| Research methods | | | | | | | | | | |
| Sample size | | | −0.001 | 0.001 | 6.49 | .219 | [−0.002, 0.001] | | | |
| Study design | | | | | | | | 1.16 | 2, 6.48 | .37 |
| RCT | 273 | 42 | 0.31 | 0.04 | 34.47 | <.01 | [0.23, 0.39] | | | |
| Quasi | 21 | 3 | 0.13 | 0.11 | 1.94 | .34 | [−0.34, 0.61] | | | |
| Other | 57 | 8 | 0.53 | 0.13 | 6.27 | .01 | [0.21, 0.84] | | | |

*Note*. PA = phonological awareness; RCT = randomized controlled trial.



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interventions not characterized as multisensory ($g$ = .0.34; $t$ = −1.41, $df$ = 3.19, $p$ = .25). Group size did not moderate intervention effects ($t$ = 1.02, $df$ = 13.32, $p$ = .33).

### *Outcome Domain Characteristics*

Intervention outcome domain was a significant moderator ($F$ = 4.60, $df$ = 3, 21.8, $p$ = .01) of intervention effects, with effects on measures of reading comprehension ($g$ = 0.26) and text reading ($g$ = 0.25) appearing smaller than those on measures of word reading/spelling ($g$ = 0.34) or PA ($g$ = 0.44).

### *Research Methods*

Neither sample size ($t$ = −1.36, $df$ = 6.49, $p$ = .22) nor study design ($F$ = 1.16, $df$ = 2, 6.48, $p$ = .37) significantly moderated intervention effects.

### *Multiple Moderator Analysis*

The covariates included in the multiple moderator analysis explained 35% of the variance in effect sizes. As is illustrated in Table 5, results indicated that only dosage ($t$ = 2.3, $df$ = 12.24, $p$ = .04) and outcome domain ($F$ = 3.83, $df$ = 3, 21.9, $p$ = .02) were statistically significant moderators when controlling for other moderators. With each additional intervention hour, effect sizes tended to increase by 0.002. Intervention effects on reading comprehension outcomes tended to be smaller than effects on word reading/spelling outcomes.

### *Publication Bias*

The modified Egger's regression test result indicated no evidence of funnel plot asymmetry ($b$ = 0.02, $df$ = 16.8, $p$ = .89).

## Discussion

The present meta-analysis differed from previous meta-analyses of reading intervention research in that it examined the effects of interventions on reading outcomes for samples of students with or at risk for word RDs, defined as scores below a given threshold on a norm-referenced measure of word reading, spelling, or skills foundational to word reading and spelling. Previous reading research meta-analyses have used norm-referenced inclusion criteria or examined the degree to which severity of RD operationalized relative to norm-referenced thresholds moderated intervention effects (e.g., Gersten et al., 2020; Suggate, 2010;

TABLE 5
Results of the Multiple Meta-Regression Moderator Analysis

| Moderator | Beta | SE | *df* | *p* | *F* | *df* | *p* |
|---|---|---|---|---|---|---|---|
| Model | | | | | 1.62 | 13, 10.3 | .22 |
| Intercept | 0.149 | 0.153 | 11.175 | .350 | | | |
| Grade (K-2) | 0.123 | 0.120 | 12.769 | .324 | | | |
| Dosage | 0.002 | 0.001 | 12.248 | .040 | | | |
| Small group | 0.042 | 0.098 | 11.289 | .679 | | | |
| Multisensory | −0.218 | 0.169 | 4.268 | .263 | | | |
| Morphology/Vocabulary component | −0.04 | 0.127 | 10.376 | .760 | | | |
| Spelling component | 0.105 | 0.056 | 7.426 | .101 | | | |
| PA component | 0.053 | 0.082 | 22.647 | .530 | | | |
| Outcome | | | | | 3.83 | 3, 21.9 | .024 |
| PA | 0.101 | 0.072 | 15.521 | .180 | | | |
| Text reading | −0.087 | 0.046 | 12.485 | .084 | | | |
| Reading comprehension | −0.080 | 0.031 | 26.487 | .017 | | | |
| Sample size | −0.001 | 0.000 | 4.906 | .180 | | | |
| Study design | | | | | 1.16 | 2, 6.48 | .371 |
| Quasi | −0.265 | 0.157 | 3.494 | .177 | | | |
| Other | 0.205 | 0.123 | 10.596 | .213 | | | |

*Note*. Intercepts represent the mean effect of interventions provided to students in Grades 3–5, with the mean dosage, provided one-on-one, not described as multisensory, with morphology or vocabulary instruction, spelling instruction, and PA instruction components on word reading/spelling outcomes in studies with the mean sample size and RCT designs.



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Wanzek et al., 2018). However, these meta-analyses included studies with participants who scored below a threshold on any type of reading measure (i.e., measures of language and comprehension, as well as of word reading). We were unable to identify any other systematic review that examined effects on reading outcomes for students who scored below a norm-referenced threshold on measures of word reading, spelling, or skills foundational to word reading or spelling.

Results from this meta-analysis of 53 reading intervention studies reveal significant, positive effects on norm-referenced reading outcomes ($g = 0.33$, 95% CI [0.25, 0.41], $p < .01$) for students with or at risk for dyslexia. Findings indicate that students in Grades K-5 who score below a screening threshold of performance at or below the 25th percentile on a standardized measure of word reading, spelling, or skills foundational to word reading and spelling are likely to benefit from reading interventions that include a word reading instructional component. Like other reviews of reading intervention research that include the full span of elementary grades (e.g., Neitzel et al., 2022), this meta-analysis included many more studies (i.e., 45 of 53) conducted with students in Grades K-2 than with students in Grades 3–5. Research suggests that reading intervention effects may be larger for younger students (e.g., Connor et al., 2013; Lovett et al., 2017). In line with these findings, the mean effect estimate for this review might have been lower had all grades levels been represented to a similar degree within included studies.

The mean intervention effect size estimated in the current review resembles mean effect sizes reported in recent meta-analyses of reading intervention research that included students with a range of reading and language difficulties (i.e., with RDs that may or may not have included word RDs) in the elementary grades. It is comparable in magnitude to mean effect sizes reported in the meta-analyses that included effects on standardized and unstandardized measures (e.g., Gersten et al., 2020; Neitzel et al., 2022; Wanzek et al., 2018) and in the meta-analysis that reported disaggregated effects on standardized measures (i.e., Wanzek et al., 2016). For example, Wanzek et al. (2016) reported that less extensive reading interventions (i.e., those with fewer than 100 sessions) for students with RDs in Grades K-3 yielded a mean effect of 0.49 on standardized foundational skills measures and 0.38 on standardized language or comprehension measures. In estimating the effects of reading interventions for students in Grades 1–3, Gersten et al. (2020) reported a mean effect of $g = 0.39$. Neitzel et al. (2022) reported a mean effect of 0.26 on standardized and unstandardized combined reading outcomes for students in Grades K-6. Wanzek et al. (2018) estimated a mean effect of extensive interventions (i.e., with 100 or more sessions) on Grade K-3 reading outcomes of 0.39.

It may seem surprising that the present review's mean effect estimate and the mean effect estimated by Wanzek et al. (2016), both of which were based on standardized measures, were not significantly smaller than those estimated in the reviews that combined effects on standardized and unstandardized measures. One expects meta-analyses that include effects on both standardized and unstandardized measures to yield somewhat larger mean effects, because use of researcher-developed measures in intervention research is often associated with larger effect sizes (Scammacca et al., 2015; Swanson et al., 1999). However, Gersten et al. (2020), Neitzel et al. (2022), and Wanzek et al. (2018) each articulated strict research design and measurement requirements within their inclusion criteria; as a result, many effect sizes included in their meta-analyses were based on standardized, norm-referenced measures (e.g., in the meta-analysis conducted by Wanzek et al., 2018, only 24 of the 328 effect sizes used were derived from researcher-developed measures).

The fact that mean effects reported in other meta-analyses were close to our mean effect estimate and frequently within the confidence interval we identified for that mean effect (i.e., between 0.25 and 0.41) reflects a few realities. First, while students with dyslexia represent the lower end of a distribution of code-based reading skills, the distribution is a continuous one. Students above our norm-referenced threshold (i.e., students among those identified as students with RDs broadly defined in previous systematic reviews) may also struggle with word reading, spelling, and skills foundational to word reading and spelling, if to a slightly lesser degree. It makes sense that they also benefit from interventions that address those areas of weakness. The similarity of our finding to those reported previously is also a reminder that students with dyslexia frequently experience difficulties with language comprehension (Adlof & Hogan, 2018; Pennington et al., 2012; Snowling & Hulme, 2021), whether because language comprehension difficulties emerge as a secondary consequence of word RDs (Stanovich, 2009), are a different manifestation of the same underlying cognitive difficulty that causes dyslexia (Tallal et al., 1997), or are distinct but commonly comorbid difficulties (Catts et al., 2005). The truth is, most students with RDs experience both code- and language-based difficulties. In separate studies conducted with third graders (Jones et al., 2016) and eighth graders (Cirino et al., 2013), more than 85% of students with reading comprehension difficulties also had difficulties with accurate or fluent word reading. Similarly, most students who have word RDs also have difficulties with vocabulary knowledge (Clemens et al., 2017) and broader listening comprehension (Capin et al., 2021). Capin et al. (2021) found that only five percent of fourth-grade students with RDs presented specific word RDs (i.e., reading profiles characterized by more significant weaknesses in word reading than listening comprehension). Thus, it may not be surprising that the effects of interventions for students with or at risk for dyslexia are similar to effects estimated in previous meta-analyses that



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included studies with participants at risk for RDs more broadly.

## Moderators of Intervention Effects

Though we explored the effects of eight moderator variables representing participant, intervention, outcome measure, and study characteristics, only dosage and outcome domain emerged as significant moderators of intervention effects in the multiple moderator model. When controlling for the effects of other moderators, higher dosage interventions, on average, yielded slightly larger effect sizes than lower dosage interventions: with each additional intervention hour, effect size tended to increase by 0.002 ($p = .040$). This is an important finding, because while practitioners are frequently advised to increase the intensity of reading interventions for students who demonstrate weak response to less-intensive interventions (Denton, 2012; Gersten et al., 2008), empirical examinations of the impact of dosage (i.e., one important dimension of intensity) on reading outcomes for primary-grade students with RDs have yielded mixed results. Some studies that directly examined the effects of intervention dosage have reported positive impacts of increased dosage (e.g., Al Otaiba et al., 2005), at least for students with the most severe RDs (Vaughn et al., 2003); other studies have found no benefit (e.g., Denton et al., 2011; Hatcher et al., 2006). A number of meta-analyses (e.g., Suggate, 2010; Wanzek et al., 2016) have reported nonsignificant relations between dosage of instruction and reading outcomes for primary-grade students with RDs, and Roberts, Hall, et al. (2021) reported a nonlinear relation between these two variables. Our finding that increased dosage is associated with accelerated reading development provides more evidence supporting recommendations to intensify interventions by increasing dosage to accelerate gains for students who are at risk for or have dyslexia. Intervention intensity may be particularly important in the context of students who have significant word RDs.

Outcome domain also moderated intervention effects. Specifically, effects on reading comprehension outcomes tended to be smaller than effects on word reading or spelling outcomes. This finding corroborates results reported by Gersten et al. (2020) and Neitzel et al. (2022), who found that outcome domain statistically significantly moderated intervention effects. Gersten et al. determined that effects on word/pseudoword measures were greater than effects on passage reading or reading comprehension measures; Neitzel et al. found that effects on alphabetics (i.e., PA, print awareness, letter naming, phonics knowledge, decoding, and encoding) and passage reading fluency measures were larger than those on general reading performance measures. Our finding, then, is similar to findings reported in previous research and further underscores the relative pliancy of word reading and spelling relative to reading comprehension. However, this finding should also be interpreted in light of our inclusion criteria, which required that interventions focus on at least one foundational skills domain but did not require that they focus on language or reading comprehension. As a result of this inclusion criterion, most included interventions primarily targeted foundational skills and knowledge, making word reading and spelling outcomes more aligned to interventions. Although improvements in foundational skills would be expected to translate to improved reading comprehension (Hoover & Gough, 1990), reading comprehension was nevertheless a less proximal outcome for this corpus of studies.

Student grade level was not a significant moderator of intervention effects. This finding is somewhat unexpected, as individual intervention studies that have manipulated the grade level at which students receive intervention suggest that earlier intervention is more effective than later intervention (e.g., Connor et al., 2013; Lovett et al., 2017). Wanzek and Vaughn (2007), in their meta-analysis of research examining the effects of extensive reading interventions on reading outcomes for K-3 students, determined that grade level significantly moderated intervention effects, with interventions delivered to younger students associated with larger effects. Additionally, mean effects reported in the one meta-analysis that focused solely on upper elementary-grade students (Donegan & Wanzek, 2021; mean $g = 0.09$ on standardized foundational skills outcomes and 0.13 on standardized measures of oral language or reading comprehension) were, descriptively speaking, smaller than those reported in recent meta-analyses of research with students in Grades 1–3 or K-3. It is worth noting that, in the single moderator analysis, studies with students in Grades 3–5 were associated with smaller effects ($g = 0.16$) than studies with students in Grades K-2 ($g = 0.36$), although this difference was not statistically significant. One explanation for the nonsignificant finding of the grade level moderator analysis is that we only identified eight studies for inclusion that examined the effects of instruction on outcomes for students in Grades 3–5, which reduced the statistical power of our analysis to detect a true effect. The lack of statistically significant grade-level effects reported in other previous meta-analyses (e.g., Gersten et al., 2020; Wanzek et al., 2016) may reflect similar constraints. Thus, there is a need for more research examining the effects of reading intervention on reading outcomes for upper elementary students with significant word RDs. When such research exists, the interaction of student grade level and intervention effects may be a variable worthy of exploration in future, better-powered meta-analyses.

Provision of (a) morphology or vocabulary instruction, (b) encoding instruction, or (c) PA instruction in addition to word reading instruction did not moderate intervention impact. Previous meta-analyses have also explored moderating effects of intervention components



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on reading outcomes, with mixed results that suggest nuance in the moderating effects of intervention components and potential interactions with participant characteristics. For example, Suggate (2010) reported that while neither grade level nor intervention component (i.e., PA, phonics, meaning-based, or mixed) variables on their own explained additional variance in the overall effect size, adding an intervention component by grade interaction term significantly improved the proportion of variance explained in the overall effect size. There appeared to be an advantage for phonics interventions from kindergarten to Grade 1, after which interventions with a comprehension component began to predominate and the advantage for phonics interventions dissipated.

Our intervention components single moderator analyses findings both corroborated and contradicted findings reported by Gersten et al. (2020). In the Gersten et al.'s study, providing instruction in encoding or writing yielded significantly higher effect sizes on reading outcomes for students in Grades 1–3. In contrast, interventions that included a PA instruction component tended to result in *smaller* effects on reading fluency outcomes. The results of our single moderator analysis provided support for the Gersten et al.'s finding about the benefit of incorporating spelling instruction into reading interventions for students with dyslexia. However, we did not find that interventions that included a PA instructional component were associated with smaller effects. The difference between this latter finding and the one reported by Gersten et al. may reflect the fact that studies included in the present meta-analysis included students with more significant word-level RDs. For this population of students, it may be beneficial (or at least may not be *less* beneficial) to provide PA instruction in addition to instruction in word reading.

The effects of interventions that were explicitly described as multisensory ($g$ = 0.20) did not differ from interventions not characterized as multisensory ($g$ = 0.34; $p$ = .25). Our analysis was underpowered, and this is certainly a variable that deserves further study. Still, it seems evident, based on this and other research (e.g., Al Otaiba et al., 2018; Stevens et al., 2020) that there is not sufficient evidence for the benefit of reading instructional programs that describe themselves as multisensory to require that school districts use them in place of other evidence-based instructional approaches (i.e., other explicit, systematic approaches to foundational skills instruction) that do not describe themselves as multisensory.

Student group size was also not a statistically significant moderator of intervention effects. Our finding corroborates findings reported in a review by Elbaum et al. (2000), which suggested that small-group and one-on-one interventions have similar effects on reading outcomes. In agreement with Elbaum et al., neither Suggate (2010) nor Wanzek et al. (2016) reported significant associations between group size and intervention effect. However, our finding differs from one reported by Ehri et al. (2007), who directly manipulated the group size variable and found that one-on-one tutoring was more effective than small-group tutoring using the same program. Donegan and Wanzek (2021), Wanzek et al. (2018), and Neitzel et al. (2022) also reported that effect sizes associated with one-on-one interventions were larger than ones associated with small-group interventions. Descriptively speaking, studies in the present review that provided instruction to small groups of students appeared to have slightly smaller effects ($g$ = 0.20) than studies that provided instruction one-on-one ($g$ = 0.30). However, this difference was not statistically significant. The group-size variable may be worthy of study in future, better-powered research meta-analyses.

Neither study sample size nor research design were significant moderators of intervention effects. This is as it should be, given that study methods should not influence the impact of the intervention the study is evaluating. Nonetheless, the finding about the non-significant effect of research design is in contrast to findings reported in previous meta-analyses that demonstrate larger effect sizes for quasi-experimental studies than for RCT designs (e.g., Suggate, 2010). Scammacca et al. (2015) attributed their finding that the effects of reading interventions declined over time in part to the use of more rigorous research designs in recent years. Similarly, research has demonstrated that sample size is often inversely associated with effect size (e.g., Cheung & Slavin, 2012; Slavin & Smith, 2009). Slavin and Smith proposed that inverse relations between sample size and effect size may be due to publication bias, either because (a) small-sample studies need to achieve much larger differences between groups to achieve statistically significant findings or (b) small-sample studies reporting null effects are less likely to be published because they are underpowered, whereas journal editors are more likely to publish large-sample studies with null effects. Alternatively, this inverse relation may be due to super-realization bias: small studies are often conducted in more controlled settings that lack ecological validity but are more likely to yield large effects (Slavin & Smith, 2009). The fact that our meta-analysis did not replicate these findings may reflect the stringent standards we imposed regarding research design and methods (e.g., we required all studies to use norm-referenced measures at screening or pre-test). The vast majority of our studies (77%) were RCTs, and study mean sample size was above 100 (M = 114.21, SD = 79.45, range = 32–422).

## Strengths and Limitations of The Present Review

There were several methodological strengths of this review. To identify the largest number of studies relevant to research questions, the research team searched for studies published during the last four decades (a longer span of time than has been searched by the author team of any relevant previous meta-analysis). The search included grey



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literature for the same reason, as well as to avoid bias in estimating intervention effects. Analyses used RVE to account for dependencies in the data and employed a multiple meta-regression model to explore the effects of each moderator variable while controlling for the potentially confounding effects of other moderators.

Relative to other meta-analyses on the topic of reading interventions, the present meta-analysis had a large sample of studies ($k$ = 53) that employed experimental or QEDs and standardized, norm-referenced reading measures. The large number of studies identified for inclusion was a welcome surprise, given our stringent criteria for inclusion (note that more than 100 studies were excluded solely because participants did not have documented word RDs according to our inclusion criterion). That said, our moderator analyses would have been better powered had our corpus of included studies been even larger. The ERIC database indexes grey literature content (e.g., research reports, curriculum and teaching guides, conference papers, dissertations, and theses) published by 1,057 selected centers, agencies, programs, associations, and non-profit organizations. Still, there are other databases (e.g., Google Scholar; ProQuest; Theses Global) that might have yielded a larger pool of grey literature reports for screening and potential inclusion. Searching more databases could have resulted in identification of additional studies reported in peer-reviewed publications as well. Finally, conducting forward and backward searches for all articles identified for inclusion in the present review could have resulted in the identification of additional studies.

The fact that our ancestral search of articles included in previous reading intervention research meta-analyses yielded 82 articles that did not appear in our database search (14 of which met our inclusion criteria) was another cause for concern. At the same time, it is useful to observe the lack of overlap in the body of studies identified for inclusion in previous reviews of reading research (see Appendix C). Our search identified several studies that were not included in previous meta-analyses (i.e., 12 of our 53 included studies [23%] did not appear in a previous meta-analysis). Of the 41 studies included in our meta-analysis that were also included in at least one of the previous meta-analyses, 13 appeared in only *one* of the other meta-analyses (i.e., the other six previous meta-analyses did not include the study); 15 appeared in only *two* previous meta-analyses.

Researchers frequently observe lack of overlap in articles identified for inclusion in systematic reviews on the same topic. For example, in concurrent replications of a meta-analysis published by Makel and Plucker (2014), Lemons et al. (2016), and Makel et al. (2016) set out to replicate a set of search procedures outlined in the original meta-analysis. However, when searching for studies published between 1997 and 2013, Lemons et al. identified 70 studies eligible for inclusion; Makel et al. identified 109 studies eligible for inclusion during the same time frame. Of the studies that Lemons et al. included in their review, 46 did not appear in the Makel et al. literature search. Conversely, 11 of the studies included by Makel et al. in their review did not appear in the Lemons et al. literature search. Lemons et al. explained that the most likely explanation for each author team's failure to identify these studies during their search "is variation in how the different databases that were searched index the targeted journals. For example, different databases vary in whether specific journals are indexed as full text or abstract only" (p. 218).

It may also have been a limitation that we only explored intervention effects on norm-referenced reading assessments. While exclusively reporting effects on norm-referenced measures enabled us to more confidently compare intervention effects across studies and participants, it also meant that we were unable to examine effects on researcher-developed measures. This prevented us from better understanding intervention effects on assessments that were more proximal to the interventions being delivered, or on assessments that focused on particular subtypes of word reading or spelling items (e.g., ones that focused on a corpus of words of a particular type).

Finally, this meta-analysis was limited by the information provided in included studies. For example, researchers frequently failed to describe the comparison condition in any level of detail. It had originally been our intent to examine whether effects on comprehension were moderated by the nature of the comparison condition [e.g., distinguishing studies for which comparison students received (a) an alternate supplemental reading intervention provided by their schools, (b) core reading instruction while treatment-group peers were pulled for an intervention that supplanted core reading instruction, and (c) no reading instruction while treatment students received the reading intervention]. Due to insufficient description of these factors in the included studies, analysis of the impact of the counterfactual was not possible.

It was also sometimes difficult to reliably code for the content components of interventions, as authors provided very little detail about the reading interventions they were testing. When only a sentence or two in each publication was devoted to describing intervention components, it felt plausible that, say, PA instruction might have been a component of an intervention even when it was not mentioned explicitly. Conversely, there were times when authors briefly mentioned a particular component but in such a way that it was unclear whether it could be described as a full, stand-alone intervention component. For example, an author team might describe drawing students' attention to word meanings when warranted; we struggled to decide if this should "count" as providing vocabulary instruction.

As a final example, lack of information provided about students' initial word reading, spelling, or foundational reading skill performance prevented us from examining the degree to which the severity of students' dyslexia moderated intervention effects. It was difficult to rank or



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categorize study samples in a meaningful way due to the fact that many included studies *either* (a) reported pretest standard scores on a measure of word reading, spelling, or foundational skills (i.e., they did not report using a norm-referenced screening threshold) *or* (b) reported using a norm-referenced screening threshold (i.e., they did not report pretest standard scores on a measure of word reading, spelling, or foundational skills). If future research could, either in the body of the manuscript or within online supplemental materials, provide more information about sample pretest performance and the nature of instruction in treatment and comparison conditions, it would aid the work of future meta-analysts.

## Implications for Policy, Practice, and Future Research

At the most basic level, findings from this meta-analysis support the provision of reading interventions that include instruction in foundational skills to students with or at risk for dyslexia in Grades K-5. Higher dosage interventions yield slightly higher effect sizes, supporting recommendations for educators to intensify interventions by increasing dosage to accelerate reading gains for students with or at risk for dyslexia.

It may be valuable for school personnel to know that effects of one-on-one interventions included in this meta-analysis did not meaningfully differ from effects achieved by small-group intervention, as small-group interventions are usually more feasible for educators and school systems to implement. That said, instructional group size may be a variable worthy of further explanation in future, better-powered meta-analyses of reading intervention research. Gersten et al. (2020) determined in their meta-analysis that group size was not a statistically significant moderator of intervention effects when considering all Grade 1–3 reading interventions; however, for Grade 1 specifically, effects were larger if the intervention was delivered to students individually rather than to small groups. The moderating influence of group size on reading outcomes may differ when considering narrower grade-level bands.

Similarly, the question as to whether intervention effectiveness differs in early elementary compared with later-elementary grades is one worthy of further research. In the present meta-analysis, mean effects of interventions delivered earlier (in Grades K-2) were not statistically significantly larger than effects of interventions delivered later (in Grades 3–5). Descriptively speaking, though, studies with students in Grades 3–5 were associated with smaller effects ($g$ = 0.16) than studies with students in Grades K-2 ($g$ = 0.36). Furthermore, regardless of differences in effects depending on grade level, interventions with an effect size of 0.33 are more likely to close the achievement gap for children in kindergarten than for children in Grade 5, as RDs tend to become more significant as children progress through the grades (e.g., Francis et al., 1996; Juel, 1988). In addition, in the absence of intervention, children with RDs are at risk for developing psychosocial and behavioral difficulties (Morgan et al., 2008). For all of these reasons, it is likely more prudent and cost-effective for school districts to intervene early than to intervene late if they are faced with a decision about how to direct limited intervention resources.

Our findings did not provide conclusive answers about the intervention components that distinguish more effective interventions from less effective ones when it comes to this population of students. That said, single moderator analysis results indicated that interventions with spelling instruction in addition to word reading instruction were more effective ($g$ = 0.37) than those without a spelling instruction component ($g$ = 0.23), a finding that was also reported by Gersten et al. (2020). Therefore, school administrators or educators selecting foundational reading interventions for elementary students with or at risk for dyslexia may want to opt for multicomponent interventions that include both word reading and spelling components. On the other hand, findings did not suggest that they should opt for interventions that describe themselves as multisensory over other evidence-based interventions that do not describe themselves in this way.

There is a final set of implications to point out, or at least a final set of questions to pose. In recent years, well-intentioned advocacy groups have pushed for the passage of laws mandating distinct approaches to instruction and assessment for students with dyslexia (Petscher et al., 2019). Gearin et al. (2021) reported that 47 U.S. states have now adopted legislation that requires creating and implementing dyslexia-specific processes for identifying and treating students. Although it may be too early to determine the effects of these policies, many new dyslexia laws promote practices shown to benefit diverse samples of students with RDs, such as universal screening, preventive intervention, explicit instruction, and use evidence-based reading programs. However, aspects of this legislative movement have been controversial (Elliott, 2020). Based on research that shows it is difficult to meaningfully distinguish between poor readers who are dyslexic and poor readers who are not dyslexic, Miciak and Fletcher (2020) have suggested that dyslexia identification and instruction need not be separated from other schoolwide, multitier systems of support that are implemented to prevent and treat RDs. There are some indications that these new laws may lead to uncertainty among educators about whether students with RDs should receive assessment and instruction through the Individuals with Disabilities Education Act (IDEA; 2006) or support through a dyslexia-specific processes outside of IDEA (Lindstrom, 2019; Tucker, 2015). This confusion has contributed to cases in which students with dyslexia who should have been referred for special education evaluation were not referred, or not referred in a timely fashion (Texas Education Agency, 2018).



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The debates about the value of the term dyslexia and issues related to classification validity (Elliott, 2020; Elliott & Grigorenko, 2014) are outside the scope of this paper. It is also worth noting that the way our inclusion criteria define dyslexia (i.e., significant word RDs, or word RDs below a given threshold) is commonly but not universally adopted. There are those who define dyslexia as *specific* word RDs (i.e., ones that are not accompanied by language comprehension difficulties), an encapsulated weakness in decoding found amid a "sea of strengths" in reasoning and creativity (Shaywitz & Shaywitz, 2020, p. 56) often described as "classic" dyslexia (Hogan et al., 2014, p. 201). The results of this meta-analysis are specific to the definition of dyslexia we articulate in our inclusion criteria. Finally, our methods do not allow us to draw conclusions about the degree to which the same interventions are associated with similar effects for these two populations of students: interventions that have been evaluated with populations of students with RDs broadly defined (i.e., interventions included in previous meta-analyses of reading intervention research) may tend to differ, perhaps in terms of content or method of delivery or intensity, from the subset of interventions evaluated in studies included in our meta-analysis (i.e., ones that were delivered to students with significant word RDs).

That said, interventions reviewed in this meta-analysis are more alike than unalike to interventions reviewed in the past. For instance, 35 of 53 included studies evaluated the effects of multicomponent interventions that focused on both code- and meaning-based elements of reading; previous systematic reviews have also observed the prevalence of multicomponent interventions among included studies (e.g., Gersten et al., 2020). Intervention dosage also appears similar to that reported in previous meta-analyses that reported this information (e.g., Denton et al., 2022; Gersten et al., 2020). The way in which the pattern of findings in the present review resembles patterns of findings reported in previous systematic reviews that evaluated the effects of interventions delivered to students with more broadly defined RDs (i.e., when considering both the magnitude of effects and the influence of moderators) suggests that similar reading interventions hold comparable levels of promise in remediating significant word RDs and in remediating RDs broadly defined. These findings should motivate future research that can answer the question: Is it necessary for students with dyslexia to receive instruction that is qualitatively different (e.g., such that it must be provided by different teachers, in different settings, using different materials) than that provided to students with broadly defined RDs?

## NOTES

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## Supporting Information

Additional supporting information may be found in the online version of this article on the publisher's website:10.1002/rrq.477/suppinfo

**Appendix A.** Comparing Searches in Previous Meta-Analyses of Reading Intervention Research

**Appendix B.** Definitions of Variables Used in Coding

**Appendix C.** Overlap Between Studies Included in the Present Review and Studies Included in Other Recent Reviews



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